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## Optical side input/output circuit and optical connector


### Abstract

An object is to provide an optical side input/output circuit that has wavelength selectivity and is easily disposed at multiple points in a transmission path, and an optical connector. An optical side input/output circuit according to the present invention includes: a grating portion in which a fiber Bragg grating that reflects light of a desired wavelength is formed in the core of an optical fiber, the light of the desired wavelength being of light propagating in the core; and a tap portion that is disposed at a stage before the grating portion in the propagation direction of the light, and is provided with a tap waveguide that outputs the light reflected by the grating portion from a side surface of the optical fiber.



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## Field of Classification Search

**USPC:** None

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## Background/Summary

## CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a 371 U.S. National Phase of International Application No. PCT/JP2020/035617, filed on Sep. 18, 2020. The entire disclosure of the above application is incorporated herein by reference.

## TECHNICAL FIELD

(2) The present disclosure relates to an optical side input/output circuit that inputs/outputs light from a side surface of an optical fiber, and an optical connector including the optical side input/output circuit.

## BACKGROUND ART

(3) As an optical branching technique, a wavelength multiplexing coupler or the like using an arrayed waveguide grating is known. Also, to realize optical sensing and monitoring of a transmission path, an optical side output technique using a tap waveguide has been suggested. By the optical side output technique, an optical waveguide is formed by laser processing in a fiber, and part of the power of light is output from the core (see Non Patent Literature 1, for example).

## CITATION LIST

### Non Patent Literature

(4) Non Patent Literature 1: Peng Ji et al, optics express, vol. 26, no. 12, p 14972-14981, (2018). Non Patent Literature 2: T. Erdogan, Journal of Lightw. Technol., vol. 15, no. 8, (1997).

## SUMMARY OF INVENTION

### Technical Problem

(5) A conventional wavelength multiplexing coupler is large in size, and has reflection and loss that are larger at connecting points. Therefore, it is difficult to dispose such wavelength multiplexing couplers at multiple points in a transmission path. Further, conventional tap waveguides are easily disposed at multiple points in a transmission path, but it is difficult to increase the wavelength selectivity of these tap waveguides.

(6) Therefore, to solve the above problem, the present invention aims to provide an optical side input/output circuit that has wavelength selectivity and is easily disposed at multiple points in a transmission path, and an optical connector.

### Solution to Problem

(7) To achieve the above object, an optical side input/output circuit according to the present invention includes a tap waveguide having wavelength selectivity.

(8) Specifically, an optical side input/output circuit according to the present invention includes: a grating portion in which a fiber Bragg grating that reflects light of a desired wavelength is formed in the core of an optical fiber, the light of the desired wavelength being of light propagating in the core; and a tap portion that is disposed at a stage before the grating portion in the propagation direction of the light, and is provided with a tap waveguide that outputs a reflected light reflected by the grating portion from a side surface of the optical fiber.

(9) Further, an optical connector according to the present invention includes the optical side input/output circuit.

(10) The optical side input/output circuit has a fiber Bragg grating formed to give wavelength selectivity to the tap waveguide. Having the tap waveguide, the optical side input/output circuit is easily disposed at multiple points in a transmission path. Further, the optical side input/output circuit can input/output light of a desired wavelength with the fiber Bragg grating. Thus, the present invention can provide an optical side input/output circuit that has wavelength selectivity and is easily disposed at multiple points in a transmission path, and an optical connector.

(11) The optical side input/output circuit according to the present invention is characterized in that a plurality of sets of the tap portion and the grating portion is continuously arranged in the optical fiber. Light of a desired wavelength can be supplied to the optical fiber.

(12) The optical side input/output circuit according to the present invention further includes a light

receiver that is disposed on the side surface of the optical fiber, and receives the reflected light output from the tap portion. Light of a desired wavelength can be received from the optical fiber.
(13) Note that the respective inventions described above can be combined as appropriate.
Advantageous Effects of Invention
(14) The present invention can provide an optical side input/output circuit that has wavelength selectivity and is easily disposed at multiple points in a transmission path, and an optical connector.

## Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a diagram for explaining an optical side input/output circuit according to the present invention.
(2) FIG. **2** is a graph for explaining the characteristics of an optical side input/output circuit according to the present invention.
(3) FIG. **3** is a diagram for explaining the characteristics of an optical side input/output circuit according to the present invention.
(4) FIG. **4** is a graph for explaining the characteristics of an optical side input/output circuit according to the present invention.
(5) FIG. **5** is a graph for explaining the characteristics of an optical side input/output circuit according to the present invention.
(6) FIG. **6** is a graph for explaining the characteristics of an optical side input/output circuit according to the present invention.
(7) FIG. **7** is a graph for explaining the characteristics of an optical side input/output circuit according to the present invention.
(8) FIG. **8** is a graph for explaining the characteristics of an optical side input/output circuit according to the present invention.
(9) FIG. **9** is a diagram for explaining an optical side input/output circuit according to the present invention.
(10) FIG. **10** is a diagram for explaining an optical side input/output circuit according to the present invention.
(11) FIG. **11** is a diagram for explaining an optical connector according to the present invention.
(12) FIG. **12** is a diagram for explaining an optical connector according to the present invention.
DESCRIPTION OF EMBODIMENTS
(13) Embodiments of the present invention are described below with reference to the accompanying drawings. The embodiments described below are examples of the present invention, and the present invention is not limited to these embodiments. Note that components having the same reference signs in the present description and the drawings indicate the same components.
First Embodiment
(14) FIG. **1** is a diagram for explaining an optical side input/output circuit **301** according to this embodiment. The optical side input/output circuit **301** includes: a grating portion **20** in which a fiber Bragg grating **21** that reflects light of a desired wavelength is formed in the core **51** of an optical fiber **50**, the light of the desired wavelength being of light propagating in the core **51**; and a tap portion **10** that is disposed at a stage before the grating portion **20** in the propagation direction of the light, and is provided with a tap waveguide **53** that outputs the light reflected by the grating portion **20** from a side surface of the optical fiber **50**.
(15) The optical fiber **50** is a step-index fiber that is defined by the diameter d.sub.c of the core **51**, the diameter df of the optical fiber **50**, the refractive index n.sub.core of the core **51**, and the refractive index n.sub.clad of a cladding **52**. In the optical fiber **50**, the tap portion **10** and the grating portion **20** are formed in this order in the longitudinal direction. The direction in which

light entering from the tap waveguide **53** travels is the optical waveguide direction. In FIG. **1**, the optical waveguide direction is the direction from left to right. Further, the direction in which the tap waveguide **53** faces the side surface of the optical fiber **50** from the core **51** is the tap direction. In FIG. **1**, the tap direction is the direction inclined in the opposite direction to the optical waveguide direction.

(16) The grating portion **20** reflects only the light of a desired wavelength in the light that has traveled in the optical waveguide direction and passed through the tap portion **10**, and returns the reflected light to the tap portion **10**. The coupling efficiency of light from the core **51** to the tap waveguide **53** in the tap portion **10** greatly depends on the light propagation direction. Specifically, light traveling in the optical waveguide direction in the core **51** is hardly coupled to the tap waveguide **53**. On the other hand, light traveling in the direction opposite to the optical waveguide direction in the core **51** can be coupled to the tap waveguide **53**, as appropriate a is set according to the mode coupling theory (see Non Patent Literature 1, for example). Here, α is an angle (on the acute angle side) formed by the tap waveguide **53** and the core **51**.

(17) The optical side input/output circuit **301** transmits light in the optical waveguide direction without being coupled to the tap waveguide **53** in the tap portion **10**, reflects only a desired wavelength in the grating portion **20**, returns the reflected light (light in the opposite direction to the optical waveguide direction) to the tap portion **10**, and couples the reflected light to the tap waveguide **53**.

(18) The tap waveguide **53** and the grating **21** can be formed by locally modulating a refractive index of the optical fiber **50**, using femtosecond laser processing, for example. Here, the amounts of refractive index modulation (differences from the refractive index before modulation) in the core and the cladding are denoted by δn.sub.core and δn.sub.clad, respectively. That is, in the case of the core **51** (the overlapping portions of the grating **21**, the core **51**, and the tap waveguide **53**), the refractive index after the modulation is expressed as n.sub.core+δn.sub.core. In the case of the cladding **52** (the tap waveguide **53** excluding the overlapping portions), the refractive index after the modulation is n.sub.clad+δn.sub.clad.

(19) FIG. **2** is a graph for explaining the α dependence of the coupling efficiency at which light propagating in the opposite direction to the light propagation direction in the core **51** is coupled to the tap waveguide **53**. A tap waveguide **53** in which δn.sub.core=δn.sub.clad=0.006, and d.sub.t=6 μm is formed for a fiber structure having a step-index refractive index distribution where d.sub.c=8.2 μm, n.sub.core=1.449081, and n.sub.clad=1.444. The solid line indicates the efficiency of coupling to the core **51** (the ratio of light that is from the grating portion **20** and is propagating in the core **51**), and the dotted line indicates the efficiency of coupling to the tap waveguide **53** (the ratio of light that is from the grating portion **20** and is coupled to the tap waveguide **53**). As can be seen from the graph, the efficiency of coupling to the tap waveguide **53** changes with α, and the coupling efficiency is maximized when α=2.5°.

(20) FIG. **3** is a diagram for explaining an example of an electrical field distribution in the tap portion **10** in a case where α=2.5°. Here, δn.sub.core=δn.sub.clad=0.006, and d.sub.t=6 μm. Reflected light is supposed to propagate from right to left in the drawing. The wavelength of light is 1550 nm. In the tap portion **10**, a state in which the light propagating in the core **51** is coupled to the tap waveguide **53** can be seen. However, the light is not coupled to a tap waveguide **53***a* extending in the opposite direction to the propagation direction. As can be seen from this drawing, the coupling to the tap waveguide **53** has great dependence (α dependence) on the orientation of the tap waveguide.

(21) FIG. **4** is a graph for explaining the d.sub.t dependence of the coupling efficiency at which light propagating in the opposite direction to the light propagation direction in the core **51** is coupled to the tap waveguide **53**. A tap waveguide **53** in δn.sub.core=δn.sub.clad=0.006 is formed for a fiber structure having a step-index refractive index distribution where d.sub.c=8.2 μm, n.sub.core=1.449081, and n.sub.clad=1.444. Here, α=3°. The solid line indicates the efficiency of

coupling to the core **51** (the ratio of light that is from the grating portion **20** and is propagating in the core **51**), and the dotted line indicates the efficiency of coupling to the tap waveguide **53** (the ratio of light that is from the grating portion **20** and is coupled to the tap waveguide **53**). As can be seen from the graph, the efficiency of coupling to the tap waveguide **53** increases with the diameter d.sub.t of the tap waveguide **53**.

(22) FIG. **5** is a graph for explaining the δn dependence of the coupling efficiency at which light propagating in the opposite direction to the light propagation direction in the core **51** is coupled to the tap waveguide **53**. A tap waveguide **53** in which d.sub.t=6 μm is formed for a fiber structure having a step-index refractive index distribution where dc=8.2 μm, ncore=1.449081, and nclad=1.444. Here, α=3°. The solid line indicates the efficiency of coupling to the core **51** (the ratio of light that is from the grating portion **20** and is propagating in the core **51**), and the dotted line indicates the efficiency of coupling to the tap waveguide **53** (the ratio of light that is from the grating portion **20** and is coupled to the tap waveguide **53**). As can be seen from the graph, the efficiency of coupling to the tap waveguide **53** changes with the refractive index modulation amount δn, and the coupling efficiency is maximized when δn.sub.core=δn.sub.clad=0.006.

(23) As can be seen from FIGS. **4** and **5**, the efficiency of coupling to the tap waveguide depends on d.sub.t and δn, and it is possible to perform coupling about 50% from the core **51** to the tap waveguide **53** by appropriately setting d.sub.t and δn.

(24) The grating portion **20** reflects only the wavelength to be extracted by the fiber Bragg grating (FBG) **21** from the light traveling in the light propagation direction in the core **51**, and returns the reflected light to the tap portion **10**. The grating pitch Λ is calculated according to the expression shown below, from the wavelength λ in vacuum and the average value n.sub.eff of the effective refractive indexes in the grating portion **20** at the wavelength λ (see Non Patent Literature 2, for example).

(25) [MathematicalExpression1]

$$= \frac{}{2n_{\mathrm{eff}}} \quad (1)$$

(26) FIG. **6** is a graph for explaining the wavelength dependence of the grating pitch of the grating portion **20**. Note that this graph shows a calculation of a structure in which a grating having a refractive index modulation amount δn of 0.003 is provided for a fiber structure having a step-index refractive index distribution where d.sub.c=8.2 μm, n.sub.core=1.449081, and n.sub.clad=1.444. For example, where a grating pitch Λ of 534 nm is set, the grating portion **20** can selectively reflect light having a wavelength of 1550 nm.

(27) FIG. **7** is a graph for explaining the wavelength dependence of the reflectivity of the grating portion **20**. Note that this graph shows the calculation result in a case where the effective refractive index n.sub.eff=1.45, the refractive index modulation amount=0.003, the grating length Lg=1 cm, and the grating pitch Λ=534 nm. Further, the reflectivity is shown as normalized output power of the backward transmission light at the end of entrance to the grating portion **20**.

(28) Note that it is possible to adjust the passband width (the wavelength range of reflected light) by changing δn and the grating length Lg. FIG. **8** is a graph for explaining the wavelength dependence of the reflectivity of the grating portion **20** when the grating length Lg is 1.5 cm. As can be seen from the graph, the wavelength range of reflected light is narrower than the waveform of the grating length Lg=1 cm in FIG. **7**.

(29) Note that the FBG can be formed by a CO2 laser or a femtosecond laser.

Second Embodiment

(30) FIG. **9** is a diagram for explaining an optical side input/output circuit **302** according to this embodiment. The optical side input/output circuit **302** differs from the optical side input/output circuit **301** illustrated in FIG. **1** in that a plurality of sets of the tap portion **10** and the grating portion **20** is continuously arranged in the optical fiber **50**.

(31) As the sets of the tap portion **10** and the grating portion **20** are arranged in the light

propagation direction, the optical side input/output circuit **302** can perform tapping (which is taking out light of a desired wavelength from the optical fiber **50**) at any position in the transmission path. Further, as illustrated in FIG. **10**, the optical side input/output circuit **302** can form a multistage optical feed system by attaching optical feed elements **30** onto output portions on the side surface.
(32) FIGS. **11** and **12** are diagrams for explaining an optical connector **350** of this embodiment. The optical connector **350** includes an optical side input/output circuit **303**. The optical side input/output circuit **303** is the same as the optical side input/output circuit **301** described with reference to FIG. **1**, except for further including a light receiver **40** that is disposed on the side surface of the optical fiber **50** and receives the reflected light output from the tap portion **10**. Reference numeral **45** indicates the coating of the optical fiber **50**.
(33) The optical connector **350** includes a ferrule **43** that has the optical side input/output circuit **303** therein, and a connector plug **44** that serves to connect to another optical connector. The shape of the connector plug **44** is of SC type, FC type, LC type, MPO type, or the like, which is widely used. By inserting the optical side input/output circuit **303** into the optical connector **350**, it is possible to easily connect to another optical fiber **50***a*, and realize optical side inputs/outputs from the optical fiber **50**.
EFFECTS
(34) In the optical side input/output circuits and the optical connector described in the first to third embodiments, wavelength selectivity is added to the optical side input/output technology, so that light of a desired wavelength and desired power are extracted in a transmission path. For example, feed light is extracted in multiple stages, sensor control is performed, and the extracted light is input to another optical fiber. Thus, path control can be performed depending on wavelength.
REFERENCE SIGNS LIST
(35) **10** tap portion **20** grating portion **21** fiber Bragg grating (FBG) **30** optical feed element **40** light receiver **43** ferrule **44** connector plug **45** coating **50**, **50***a* optical fiber **51** core **52** cladding **53** tap waveguide **301** to **303** optical side input/output circuit **350** optical connector

## Claims

1. An optical side input/output circuit comprising: a grating portion in which a fiber Bragg grating that reflects light of a desired wavelength is formed in a core of a step-index optical fiber, the light of the desired wavelength being of light propagating in the core; and a tap portion that is disposed at a stage before the grating portion in a propagation direction of the light, and is provided with a tap waveguide that outputs a reflected light reflected by the grating portion from a side surface of the optical fiber; wherein the tap portion is configured such that the tap waveguide extends from outside the optical fiber, passes through a cladding of the optical fiber, and connects to the core, and wherein an angle formed between the core and the tap waveguide on a side opposite to the grating portion is an acute angle.
2. The optical side input/output circuit according to claim 1, wherein a plurality of sets of the tap portion and the grating portion is continuously arranged in the optical fiber.
3. The optical side input/output circuit according to claim 1, further comprising a light receiver that is disposed on the side surface of the optical fiber, and receives the reflected light output from the tap portion.
4. An optical connector comprising the optical side input/output circuit according to claim 1.